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Living Wetland Foundations: a nature-based architectural model for maintaining groundwater regimes beneath wooden pile buildings in tropical swamplands

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Abstract

Rapid urbanization in tropical wetland cities has increasingly disrupted natural groundwater regimes through land filling, soil sealing, and conventional drainage practices, resulting in the gradual deterioration of wooden pile foundations that depend on permanently saturated conditions for long-term durability. Existing studies have primarily examined wooden foundations from structural and geotechnical perspectives, while the ecological relationship between foundation performance and wetland hydrology remains insufficiently explored. This study develops a conceptual architectural model termed Living Wetland Foundation, a nature-based approach that integrates wooden pile foundations, ecological drainage systems, rainwater infiltration, and wetland hydrological processes into a unified design framework. Using a qualitative-conceptual methodology combining literature review, field observations in tropical swamplands, three-dimensional conceptual modeling, and environmental simulation, the study investigates how hydrological continuity beneath buildings can be maintained despite urban development pressures. The proposed model is founded on three principles: maintaining water-saturated conditions around wooden piles, directing rainwater infiltration toward foundation zones, and preserving ecological interactions between buildings and surrounding wetlands. The novelty of this research lies in redefining building foundations as active ecological infrastructures rather than passive structural elements. The Living Wetland Foundation model offers a new paradigm for sustainable wetland architecture by simultaneously enhancing foundation durability, groundwater recharge, and ecosystem resilience, providing a transferable design framework for tropical wetland cities facing rapid urban transformation.

Keywords Living Wetland Foundation, Wetland architecture, Nature-based infrastructure, Wooden pile foundations, Groundwater management, Tropical swampland settlements

Introduction

Wetlands are among the most productive and ecologically significant landscapes in the world. These ecosystems provide a variety of vital environmental services, including flood control, groundwater recharge, carbon sequestration, water purification, biodiversity conservation, and adaptation to climate change [1]. In tropical regions, wetlands also serve as sites for human settlements that have developed through a long-term process of adaptation to dynamic hydrological conditions. Unlike settlements that develop on relatively stable dry land, wetland settlements require architectural solutions capable of accommodating periodic flooding, soft soil conditions, and fluctuations in the water table. Consequently, vernacular architecture in wetland areas has developed unique foundation systems that not only support the building's load but also coexist with natural water processes.

Banjarmasin, located in South Kalimantan, Indonesia, is one of the largest tropical wetland cities in Southeast Asia. Historically, the city developed within a vast wetland ecosystem characterized by high groundwater levels, tidal influences, and soft alluvial soil. Traditional settlements evolved through adaptive architectural strategies that minimize environmental disruption while maintaining hydrological continuity beneath the structures. One of the most prominent construction techniques is the use of wooden pile foundations, particularly those utilizing cajuput wood (*Melaleuca cajuputi*) and various other local timber species [2]. Wooden pile foundations are among the oldest foundation techniques used in building construction. Their main advantages are their ability to effectively support loads on unstable or soft ground, such as swamps, as well as a relatively faster and more cost-effective installation process compared to concrete or steel foundations. Galam wood, in particular, is known for its exceptional strength and durability when submerged in water—a key characteristic that prevents biological degradation [3]. The architectural logic underlying the use of wooden pile foundations in tropical wetlands is very different from conventional foundation systems developed for dryland environments. Rather than attempting to remove water from the site, traditional wetland architecture accepts the presence of water as an integral part of the environmental system. Buildings are constructed on stilts so that swamp water can continue to flow beneath them, reducing hydrostatic pressure and minimizing direct contact between living spaces and standing water. This stilt house concept also helps prevent moisture problems inside the building and minimizes the risk of damage caused by wetland conditions, while allowing rainwater and swamp groundwater to flow freely beneath the house.

In this context, foundations serve not only as structural elements but also as part of a broader ecological adaptation strategy. However, rapid urbanization and the development of modern housing have significantly altered the long-standing relationship between buildings and wetland ecosystems. In recent decades, increasingly large-scale land conversion has transformed wetland landscapes into densely populated residential areas. Modern development practices generally rely on large-scale land filling, site leveling, the use of rigid paving materials, and extensive soil sealing [4]. Although these interventions facilitate urban expansion, they simultaneously disrupt the natural hydrological processes that have long sustained wetland ecosystems. The impact of urbanization on wetlands has proven to be highly significant, with studies showing that urbanization is the primary cause of wetland loss in various regions [5].

One of the most significant consequences of this change is the gradual isolation of the wooden pile foundations from the swamp environment that has long supported their durability. Under traditional conditions, galam wood foundations were always submerged in water-saturated soil. Under these conditions, the biological degradation of wood proceeds very slowly due to the limited availability of oxygen required by wood-destroying organisms. The durability of galam wood in consistently wet environments has long been recognized in local construction practices; in fact, many wooden foundations have lasted for decades to centuries as long as groundwater levels remain stable [6, 7]. Recent research indicates that erosion-causing bacteria are the primary degraders of wood in water-saturated environments, and this degradation proceeds very slowly when the wood remains submerged below the water Tables [8, 9]. Problems begin to arise when urban development alters the groundwater regime around the foundations. Land-filling practices commonly carried out in residential areas—such as filling in yards, neighborhood roads, and parking lots—create dry areas that sever the natural connection between the foundation and the swamp water. As a result, rainwater infiltration decreases, groundwater recharge declines, and foundation areas that were previously constantly saturated gradually dry out. While these dry soil conditions may facilitate construction, they actually cause galam wood piles to deteriorate or rot easily because their strength depends on being submerged in water. The main drawback of wood pile foundations is their susceptibility to damp environmental conditions, which can cause the wood to rot and deteriorate over time; therefore, selecting the right type of wood and applying proper wood treatment during construction are critical factors.

These environmental changes have impacts that go beyond mere hydrological shifts. When galam wood foundations are no longer submerged, the wood becomes vulnerable to decay, fungal attack, and other forms of biological degradation. The shift from water-saturated anaerobic conditions to a recurring wet-dry cycle accelerates wood decay and significantly reduces the service life of the foundations [6, 7]. Numerous cases of foundation failure in wetland settlements indicate that the loss of water-saturated conditions is one of the primary causes of weakened foundation structures and reduced overall building stability [10]. Comprehensive research on traditional wooden foundations in Amsterdam and Venice shows that maintaining water-saturated conditions is key to long-term preservation [9]. The approaches traditionally used in foundation engineering for tropical wetlands generally focus on structural performance, soil bearing capacity, and soft soil improvement techniques. Various studies have emphasized foundation dimensions, pile configurations, material strength, and geotechnical performance in soft soil conditions. Although these studies have made important contributions to the technical understanding of construction, they often overlook the ecological context in which the foundations operate. In particular, the relationship between foundation performance and long-term hydrological conditions has received relatively little attention.

On the other hand, environmental research on wetlands generally places greater emphasis on ecosystem restoration, biodiversity conservation, water quality improvement, and landscape management. Although such research acknowledges the impact of development on wetland ecosystems, these studies rarely address how architectural foundation design can play an active role in maintaining ecological functions. As a result, architecture and environmental management have developed as two relatively separate disciplines, even though both share a common concern for the sustainability of

wetland areas. This situation has become increasingly problematic in tropical wetland cities experiencing rapid urban growth. The challenge is no longer merely to construct structures capable of standing on soft ground. The more fundamental challenge is how to design architectural systems capable of ensuring structural stability while maintaining the environmental conditions necessary to preserve the long-term resilience of the foundations. Such an approach requires a paradigm shift from a conventional engineering perspective toward a more integrated understanding of buildings as part of a broader ecological system.

Recent developments in sustainable architecture emphasize the importance of implementing Nature-Based Solutions (NbS) to address various urban environmental challenges. Nature-based solutions harness ecological processes to improve environmental performance while providing social and economic benefits [11, 12]. In urban water management, this approach is implemented through various strategies such as green infrastructure, constructed wetlands, bioretention systems, and permeable surfaces. These systems are increasingly recognized as effective tools for reducing surface runoff, enhancing groundwater recharge, improving water quality, and mitigating flood risks [13–18]. Although drainage concepts have been widely applied in urban landscape design to manage stormwater and improve the environment, their integration into building foundation systems remains very limited. Existing applications generally focus on runoff management at the site scale and are not specifically designed to meet the hydrological needs of the foundation environment. However, drainage concepts are one of the solutions used to manage stormwater and improve the environment as a whole.

The theory of constructed wetlands provides an important foundation for understanding this integration. A constructed wetland is an environmental engineering system designed to mimic the natural functions of wetlands in managing water flow and improving environmental quality [19, 20]. Adapting these principles opens up new opportunities for more sustainable development in wetland areas. Research on constructed wetlands shows that these systems are capable of treating various types of wastewater while maintaining important ecological functions [20]. Furthermore, advancements in environmental engineering and smart water management highlight the importance of integrating hydrological monitoring into infrastructure design. These developments suggest that future foundation systems will not only serve as supporting structures but also as components of environmental management capable of responding to changes in groundwater conditions.

This perspective aligns with trends in ecological architecture that increasingly emphasize the integration of environmental performance into building systems. The significance of this issue is not limited to Banjarmasin. Across Southeast Asia, wetland areas face similar pressures due to urbanization, land reclamation, and environmental changes. Cities developing in river deltas, floodplains, peatlands, and coastal areas face growing challenges related to hydrological disturbances and foundation stability [21, 22]. Research in Ho Chi Minh City shows that urban infrastructure expansion has led to the loss of 69% of wetlands and an increase in ground surface temperature ranging from 3.44 °C to 9.35 °C over the past three decades [22]. Studies on wetland restoration in Indonesia highlight the importance of a holistic approach that integrates conservation with sustainable development [10]. Therefore, the development of architectural models capable

of harmonizing construction needs with the ecological processes of wetlands is a critical research agenda for sustainable urban development.

Although recognition of Nature-Based Solutions and ecological infrastructure continues to grow, research that specifically integrates these concepts into building foundation systems remains very limited. The existing literature tends to separate discussions of building foundations from environmental management. This study aims to bridge this gap by introducing an architectural model that integrates wooden pile foundations, drainage principles, and wetland hydrology into a single, cohesive conceptual framework. Research on galam wood demonstrates significant potential for further development in sustainable foundation applications [23]. The concept proposed in this study is termed the Living Wetland Foundation. This term reflects a shift in perspective from viewing foundations as isolated structural systems to viewing them as active interfaces between buildings and wetland ecosystems. The Living Wetland Foundation is designed to maintain hydrological continuity beneath the building through groundwater management, rainwater infiltration, and integrated ecological drainage strategies. In this model, the foundation not only structurally supports the building but also plays a role in maintaining environmental processes that are critical to its own resilience.

The conceptual basis of this approach is built on three interrelated main principles. The first principle is to maintain water-saturated conditions around the wooden foundation to prevent material degradation, given that galam wood requires submersion in water to ensure its durability [2, 3, 9, 10, 16, 23]. The second principle is to direct rainwater infiltration toward the foundation zone through a nature-based drainage system to maintain the groundwater regime [13, 16–18, 24]. The third principle is to preserve the ecological interaction between the building and the surrounding wetland environment, rather than isolating the building from these environmental processes [1, 11, 12, 14, 15, 20, 25]. Based on this background, this study aims to develop a conceptual model of the Living Wetland Foundation for tropical wetland environments. Specifically, this study examines how ecological drainage systems and strategies can be integrated with wooden pile foundations to maintain the groundwater regime beneath the building. Additionally, this study evaluates the environmental implications of foundation isolation resulting from landfilling practices and proposes an alternative design approach capable of restoring hydrological connectivity between the building and the wetland ecosystem.

Methods

The research method used in this study conceptually combines qualitative and quantitative approaches to obtain a comprehensive picture of the effectiveness of using galam wood, wood pile foundation, and drainage concepts in swampland. The research consisted of several main stages: literature survey, field case study, computer simulation, and data analysis. This approach allowed researchers to examine the various technical and environmental aspects that influence the implementation of this technology, as well as its impact on the stability of the swamp soil structure and ecosystem.

The first step in this research method was to conduct an in-depth literature survey. This involved reviewing scientific journals, books, research reports and other sources relevant to the research topic. The aim was to identify theories, methods, and previous research results that could support or direct this study. The literature reviewed included topics such as the use of galam wood in construction, wood pile foundation techniques,

and construction challenges in swampland. This literature survey will provide a strong theoretical basis and help in designing more focused experiments and analysis.

Field case studies are the next stage, where sites that use galam wood, and wooden pole foundations, in swampy areas are selected for direct observation. Locations such as Banjarmasin that will implement these concepts will be the main focus. In this stage, data will be collected through direct observation, interviews with construction experts and local residents, and concept testing. Direct observation will provide insight into the actual conditions on the ground, the challenges faced, and how to overcome them. Interviews with experts and local residents will provide practical and historical perspectives on the effectiveness and sustainability of the methods used.

3D (three-dimensional) experiments were conceptually conducted to test the concept of strength, stability, and durability of galam wood and wood pile foundation materials in conditions that simulate swamp soil environments. The test is in the form of identification through observation and is presented in photos of the actual condition of the foundation. This experiment is conducted by observing the conditions taken from the field case study location to ensure the suitability of the results with the real conditions in the field. The results of these conceptual experiments will provide important quantitative data to assess the performance of the materials and technologies used.

Computer simulation was used to model the condition of wooden pile foundation in swampland under various environmental conditions. The simulation uses engineering software to conceptually simulate foundation conditions, structural loads, moisture distribution, and water flow in the soil. The simulation allows researchers to test various scenarios and hypotheses that may not be directly testable in the field or laboratory. Simulations are conducted by combining concepts and field observations. The simulation results will provide additional insights into the potential and benefits of using galam wood, wood pile foundations, and assist in designing effective drainage strategies.

The final stage of this research method is the analysis of data collected from literature surveys, field case studies, concept experiments, and computer simulations. Data analysis is conducted using software techniques to identify drainage patterns, and the correlation between the foundation and the swamp soil environment. Qualitative data from interviews and observations will be analyzed using thematic analysis methods. The results of this analysis will be used to develop practical and theoretical recommendations regarding the use of galam wood, wood pile foundations, in foundation construction on swamplands.

This comprehensive research method aims to provide a comprehensive overview of the effectiveness and sustainability of wood pillar foundation in swamplands. By combining literature surveys, field case studies, concept experiments, computer simulations, and data analysis, the research is expected to produce robust and reliable findings. The findings will not only contribute to construction science and engineering, but also to sustainable and environmentally friendly development practices in swampy areas.

Results

Wood pillar foundation is one of the oldest foundation techniques used in building construction. Basically, this foundation consists of wood pillars that are embedded into the ground until they reach the hard soil layer or with a floating system (building on stilts). These wooden poles serve to transfer the load of the building to the more stable soil

layer below the surface. The use of durable wood (ulin wood) as the main material in this foundation is very popular for construction in swampland.

The main advantage of wood pillar foundation is their ability to withstand loads well on unstable or soft soils, such as swamps. In addition, the installation process of timber piles is relatively faster and cheaper than concrete or steel foundations. However, the main disadvantage of this foundation is its resistance to humid environmental conditions, which can cause the wood to weather and deteriorate over time. Therefore, the selection of the right type of wood and the treatment of wood construction are important factors in the use of this foundation.

The construction drawing of a wooden pile foundation shows an arrangement of wooden poles that are vertically embedded into the ground. Each pile is connected to horizontal beams that form a load-bearing framework. This framework is then connected to the building structure above. In addition, detailed construction drawings In modern projects, wood pillar foundation are often equipped with additional technologies such as a combination of timber and concrete materials for protection against groundwater which can extend the service life of the foundation.

The concept of a house on stilts (Fig. 1) on a swamp has long been known and used by people in the swamplands of Banjarmasin, which are characterized by being on soft soil with high swamp water. This stilt house construction is an effort to raise the floor of the building so that it does not experience flooding. The obstacle is that the swampland has unstable soil conditions. The house on stilts is built by raising the main building on poles stuck in the swamp soil. The main purpose of this design is to protect the house from waterlogging, flooding, and high humidity.

One of the main advantages of the stilt house concept is that it helps prevent moisture problems in the building (space). In addition, by raising the house above ground level, the risk of damage from swampland is minimized. The stilt structure also allows rainwater and swamp groundwater to flow smoothly underneath the house. The foundation matches the concept of lightweight building and floating reduces hydrostatic pressure on the foundation.

In addition, the materials used for poles and other supporting structures have evolved, with the use of materials that are more resistant to harsh environmental conditions. Construction drawings of modern stilt houses usually show a more complex design, with the integration of new technologies and materials such as concrete that increases the durability of the bearing capacity of the foundation that bears the greater weight of the building.

Foundation conditions under swamp water (Fig. 2) present their own challenges in building construction. Swamps are environments with high water content and soils that



Fig. 1 Foundations in swampland



Fig. 2 Foundations in waterlogged areas (River)

are often soft or unstable. The bearing capacity of swamp soils is very low. Foundations built on swamp soils must be designed for these conditions and ensure the stability and safety of the building structure above. One commonly used method is wood pillar foundation, where long poles are driven into the swamp soil until they reach the harder soil layers below.

High humidity and constant waterlogging can cause wood or other materials used in foundations to degrade faster. Therefore, proper material selection and special treatment of the material is very important. Galam wood is used as piling to strengthen the bearing capacity of the soil. Galam wood is durable and long lasting when submerged in water. Galam bark has the strength property of adhesion (friction) with swamp soil.

In foundation drawings for buildings on swamps, details of the depth and placement of galam piles are usually shown. In addition, the foundation design must also take into account the lateral pressure of the swamp soil and surrounding water. An in-depth geotechnical analysis is required to ensure that the foundation can withstand building loads and environmental conditions throughout the life of the building.

In addition to the foundation, another construction that is often carried out is the backfill around the building. Such as yard backfill that serves to provide pavement area. The foundation and backfill are important components in building construction. A strong and stable foundation is key to ensuring that the building can last for a long period of time and safely. Yard backfill, on the other hand, serves to provide a level surface and support good drainage around the building. What often happens, however, is that this part of the backfill leaves the foundation beneath the building isolated.

The foundation section includes various elements such as galam piles, foundation beams and foundation slabs. Galam piles and wooden poles are used to transfer the weight of the building to more stable soil layers. Foundation beams ('sunduk' and 'kalang') are usually placed on top of the galam piles and serve as the main support of the building structure. The foundation plate, if used, serves to distribute the load evenly across the foundation area. Foundation drawings usually show details of each of these elements, including their size, material and method of fastening.



Fig. 3 Page layout



Fig. 4 Foundation and swamp soil environment

The backfill around the foundation (Fig. 3) serves to stabilize the soil and prevent erosion. The material used for the backfill can be soil, gravel, or other materials. However, the result of this backfill has the disadvantage of impairing drainage properties. The backfill must be done carefully to ensure that there are no gaps or voids that can cause land subsidence in the future. The construction used is in the form of soil retaining sirring. And it is this soil retaining construction that makes the foundation isolated. Supposedly, the slope of the embankment should be set such that rainwater can flow away from the embankment, reducing the risk of standing water around the building.

Backfilling of the environment around the foundation is also often done. Not only the courtyard is filled in, but because the distance between the buildings in the garden is narrow, almost every side of the building is filled in (Fig. 4). This causes the foundation to be very isolated from the swamp water environment. This backfilling, with the addition of materials such as soil, sand or gravel, causes the conditions around the foundation to dry out. The actual backfill (Fig. 5) for the water catchment area should be considered to improve the drainage system around the foundation, reducing the risk of waterlogging and erosion.

The backfill process must be carried out taking into account various factors, including the type of material used, the thickness of the backfill, and the slope of the surface.



Fig. 5 Environmental fill



Fig. 6 Foundation condition with dry areas

The material used should have good drainage properties and not be prone to erosion. In addition, the thickness of the backfill should be sufficient to withstand the weight of the building and prevent future subsidence. The slope of the backfill surface should be set in such a way that rainwater can be channeled so that it does not cause puddles around the building (Fig. 6).

Backfill drawings show details of the placement of the backfill material, the thickness of the layers, and the drainage system used. In addition, the drawings also include information regarding the method of soil compaction to ensure that the backfill does not experience settlement or shifting in the future. This compaction process is very important to ensure that the backfill can withstand the weight of the building well and does not experience structural problems in the future.

The condition of the foundation under a building is very important in determining the strength and stability of a building. A strong and stable foundation is key to ensuring that a building can last for a long period of time without experiencing settlement or structural damage. Foundation planning and construction must be done carefully and cautiously, taking into account various factors such as soil type, environmental conditions, and building loads.

The lower foundation of a building usually consists of various elements such as galam piles, foundation beams ('sunduk' and 'kalang'), and foundation slabs. Each of these elements has different functions and characteristics, and must be designed in such a way that they work together to bear the weight of the building. The galam piles function to transmit the load to the more stable soil layers below the surface. Foundation beams and foundation slabs, on the other hand, serve to distribute the load evenly and resist lateral pressure from the surrounding soil.

Drawings of the foundation under a building usually show details of each foundation element, including its size, material, and method of fastening. It also includes information on the soil conditions surrounding the foundation, as well as any additional protection used to prevent damage from groundwater or other environmental factors. An

in-depth geotechnical analysis is necessary to ensure that the foundation can withstand building loads and environmental conditions throughout the life of the building.

Foundation conditions with dry soil (Fig. 6) have their own challenges and advantages. Dry soils tend to be more stable than wet soils or soils under standing water. However, dry soils can also experience shrinkage or cracks due to changes in moisture (water content), which can affect the stability of the foundation and the building above it. Therefore, the planning and construction of foundations on dry soils must be done by taking into account the properties of the soil.

One of the main advantages of dry-land foundations is the ease of construction. Foundations in dry soil are usually easier to execute, so the process of installing the foundation can be done more quickly and efficiently. In addition, the risk of damage due to groundwater or high humidity is also lower compared to foundations on wet soils. However, it should be noted that dry soils that are too loose or sandy may require additional reinforcement techniques to ensure foundation stability. The problem that will occur is that the galam wood piles in dry soil will become easily damaged (rotten). This is because the strength characteristics of galam wood must always be submerged in water.

In foundation drawings for dry soils, details are usually shown regarding the method of excavation, placement of reinforcing materials, as well as the drainage system to manage rainwater. It also includes information on protection against cracks and soil shrinkage. The use of materials such as reinforced concrete or piles can help improve foundation stability in dry soils. Geotechnical analysis is also very important to ensure that the foundation can withstand building loads and environmental conditions throughout the life of the building.

Schematic concepts (Fig. 7) for buildings located next to swamps that are still inundated with swamp water require careful planning and in-depth detailing. The scheme must take into account the soggy and watery soil conditions and their impact on building stability. In addition, the design must include solutions to manage waterlogging and ensure that the foundation is not affected by changes in the water level in the swamp.

One of the approaches used in this schematic concept is to use a siring in the form of saplings embedded into the ground until it reaches a more stable layer. Ground anchors can also be made of various materials such as wood, steel, or concrete. The choice of material depends on the environmental conditions and the backfill load to be supported. The scheme should also include an effective drainage system to drain water from around the foundation and prevent stagnant water that could damage the foundation.



Fig. 7 Conceptual scheme of the building

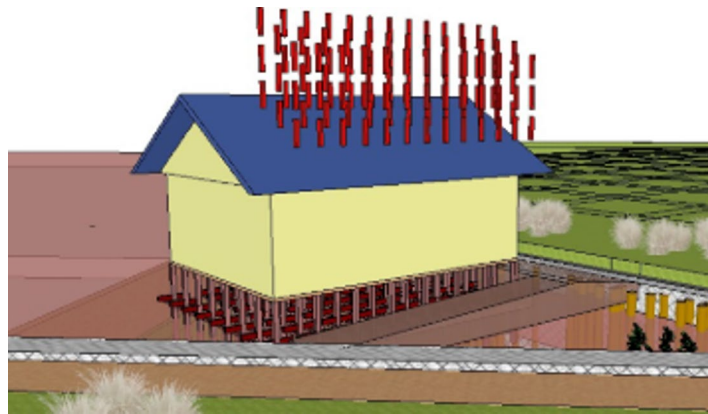


Fig. 8 Environmental and rainwater scheme

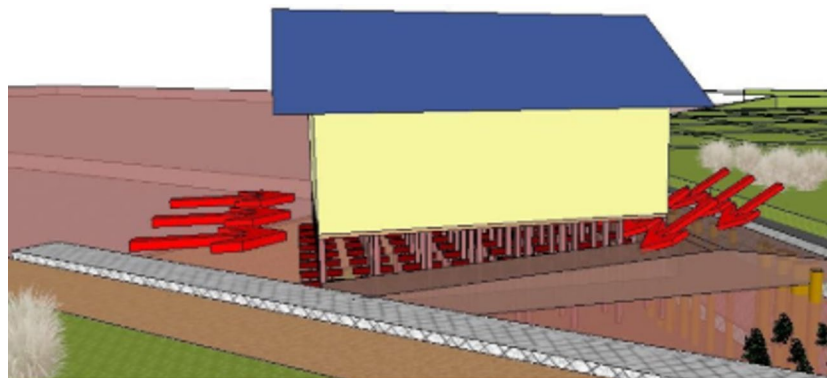


Fig. 9 Drainage scheme

Figure 8 shows the placement of the piles, the depth of the piles, and details of the drainage system and water protection. In addition, the scheme must also consider aspects of the surrounding environment, such as vegetation and the swamp's ground-water conditions. Integration between the building design and the surrounding environment is essential to ensure that the development does not damage the swamp ecosystem and remains functional over a long period of time.

The schematic of a rainy environment in Figs. 7 and 8 shows how rainwater is managed around a building. The scheme includes elements such as drainage systems, and other landscape features designed to manage stormwater flow and prevent ponding or erosion. In this scheme, rainwater is directed through drainage channels to a detention or infiltration area, under the foundation. In this case the water is directed down the building towards the foundation area. The concept is to keep the foundation waterlogged (Fig. 9).

The schematic drawing of the rainy environment shows details regarding the placement and sizing of the drainage elements, as well as the relationship between the building and the drainage system. In addition, the scheme also includes information on the care and maintenance of the drainage system to ensure that everything functions properly in the long term. Requiring proper planning, the scheme should help create a safer and more comfortable environment, as well as protect the building from rainwater damage.

One important element in this schematic is an effective drainage system. These systems typically consist of drains (pipes), culverts, and detention ponds designed to capture and convey stormwater. In addition, vegetation and water-absorbing surfaces such as paving blocks or gravel can also be used to increase water infiltration into the soil. By directing rainwater to areas under buildings, the risk of waterlogging and erosion can be significantly reduced.

Model simulation is used in the planning and design of building foundations and their environments, especially when dealing with complex environmental conditions such as swamps or wetlands. From computer simulations, multiple scenarios and conditions can be analyzed to determine the best design and most effective solution. These simulations usually include structural, hydrological and geotechnical analysis to ensure that foundations and buildings can withstand diverse environmental conditions.

The model simulation demonstrated the ability to test various designs and materials without having to conduct expensive and time-consuming physical trials. This includes simulations to determine the optimal depth of piles, and soil reinforcement and backfill methods. Identification of potential problems such as subsidence or foundation conditions will be predicted early, so that preventive measures can be taken before construction begins.

Images from model simulations (Figs. 10 and 11) show the results of analysis and predictions, such as pressure distribution in the foundation, water flow around the building, and changes in soil conditions over time. In addition, simulations can also include three-dimensional visualizations that help in understanding the interactions between various design elements. With the information obtained from simulations, planners and engineers can make better decisions and reduce the risk of errors in construction.

Underfoundation drainage is a technique used to manage groundwater (rainwater) and maintain watery conditions around building foundations. The system usually consists of pipes or channels installed under the foundation to convey water to a detention or



Fig. 10 Foundation concept with puddles



Fig. 11 Drainage concept for the lower foundation of the building

infiltration area. The main purpose of this drainage conveyance is to reduce the hydrostatic pressure on the fill and channel it under the building (foundation).

One commonly used method of drainage is the installation of perforated pipes around foundations and land fill. These pipes serve to capture groundwater (rain) and drain it to lower areas or to the main drainage system. In addition, the use of infiltration materials such as gravel or sand around the pipes can increase the effectiveness of the drainage system by allowing water to drain more easily. These systems are particularly important in wet or watery soil conditions, such as in swampy areas or land prone to flooding.

Drainage system drawings usually show the placement of pipes, the size and type of material used, and the route of water flow from the foundation to the storage area. In addition, the drawings also include information on installation methods and pipe protection to prevent blockages or damage. With a well-designed drainage system, the risk of waterlogging and foundation damage can be significantly reduced, ensuring that the building can withstand diverse environmental conditions.

Timber pile foundation is a foundation construction technique that uses wooden poles planted into the ground until it reaches the hard soil layer. Its advantages include the ability to withstand loads in soft soil and a quick and inexpensive installation process. However, its weakness is resistance to moisture which can cause weathering (deterioration) of the wood.

A house on stilts is a building design that is raised on poles to protect against flooding and moisture. Foundations under swamp water require a design that can withstand high humidity and soft soil. The general method is to use galam piling foundations that reach the hard soil layer.

A strong and stable foundation is the key to a durable building on swampland. Foundation parts include galam piles, foundation beams and foundation slabs. Yard backfill helps stabilize the soil and supports good drainage. The backfill material should have good drainage properties and be compacted to prevent subsidence.

Building design next to swamps requires detailed planning to deal with soft and watery soils. Galam piling is used to achieve a stable soil layer, with an effective drainage system to drain water and prevent inundation. The scheme should also consider aspects of the surrounding environment such as vegetation and swampland conditions.

The environmental fill around the foundation is an important factor to ensure the condition and stability of the building foundation. The backfill material should have good drainage properties and not easily eroded. Material placement and drainage system should be designed to drain water towards the foundation area.

Foundations on dry soils have challenges such as soil shrinkage or cracks. The main advantages are ease of construction and low risk of damage from moisture. The use of materials such as reinforced concrete or piles can improve stability.

Model simulation is used in building planning and design, allowing analysis of multiple scenarios without physical trials. Simulations can determine the best design and materials, and identify potential problems such as foundation conditions and ground settlement.

Drainage conveyance is a technique to manage swampy groundwater and maintain standing water around the foundation. Perforated pipes are used to capture and drain groundwater to the area below the building.

Conclusion

Drainage concept is an innovative and effective solution to manage rainwater and improve environmental quality around buildings. Its function to absorb rainwater, reduce surface flow, and increase water infiltration into the soil. By using vegetation that is resistant to wet conditions, its can reduce the risk of inundation.

To keep the foundation waterlogged, the use of wooden pole foundations and galam piles is an effective solution. These foundations are buried into the harder soil layers, providing stability to the building despite the soft and unstable nature of the swamp soil. The use of moisture-resistant materials, such as preservative-treated timber or reinforced concrete, is essential to ensure the longevity of the foundation.

Drainage of rainwater overflow to the underside of the building also requires a well-designed system. The installation of perforated pipes or channels under the foundation serves to capture groundwater and drain it to a holding area or main drainage system. This reduces hydrostatic pressure on the foundation and prevents damage from standing water. This system ensures that rainwater can flow smoothly under the building, maintaining the stability and safety of the structure above.

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Authors' contributions

NA developed the formulation, carried out the experiments, supervised the entire research, conceptualized the research, conducted the data collection, interpreted the data analyses, interpreted the result, reviewed the findings and edited the manuscript. GNS reviewed the research findings, edited, and contributed to the manuscript. AS conducted the data collection, interpreted the data analyses, and edited the manuscript. RN interpreted the data analyses, and interpreted the result. NA conceptualized the research, conducted the data collection, interpreted the data analyses, and edited the manuscript. All authors have read and approved the final manuscript.

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Declaration.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

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